Prototropic reactions o 3-phenyl and 3-hetaryl chromone derivatives in the ground and excited states.

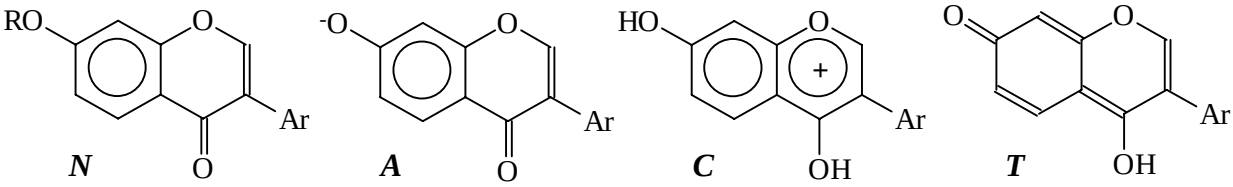
M.I. L'ovskaya[1], A.D. Roshal[1], V.P. Khilya[2]

[1] The Institute of Chemistry, Kharkiv V.N. Karazin National University, Kharkiv, Ukraine, email: alexandre.d.rochal@univer.kharkov.ua
[2] Organic Chemistry Department, Kiev National University, Kiev, Ukraine.

3-phenylchromones (isoflavones) are widespread nalural substances known as phytoestrogens and phytoalexines. Their hetaryl-analogs, which have been synthesized recently, possess interesting biological activity and may be used as medicaments. However, using these substances in pharmaceutical industry is impossible without sensitive fluorescent methods of their analysis, and, hence, without investigations of their spectral properties.

In our work hydroxyisoflavones and their thiazolyl analogs have been investigated (structure ***N***, where R – H or CH_3, Ar – $C_6H_4R_1$ or thyazole, R_1 – H or OCH_3).

Analysis of absorption and fluorescence spectra in acidity interval from H_0 ~ - 9 to pH ~ 12, as well as in solvents of different proton-donating ability, showed that prototropic reactions of chromones in the excited state differ substantially from the ones in ground state. In the case when R = H (7-hydroxychromones) fluorescence of anionic (***A***), cationic (***C***) and phototautomer (***T***) forms was observed.

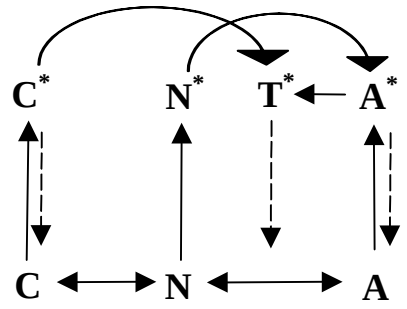


Emission of neutral form was not found. This phenomenon may be explained by presence of forbidden n-π electronic transition (~370-380 nm) possessing lower energy than the first π-π transition (305-310 nm).

Acid-base titration of chromones in ethanol-water solutions and following estimation of their pK_a and pK_b in the ground and excited states have showed that excitation leads to growth of carbonyl fragment basicity and to substantial increase of 7-hydroxygroup acidity (till 8 orders of magnitude). Consequently, presence of anion and tautomer fluorescence in large interval of pH may be explained by photodissociation process. Decrease of proton-donating solvent polarity leads to anion fluorescence quenching, but does not almost influence on phototautomer emission. Consequently, photodissociation in non-polar media results, mainly, in quasi-intramolecular proton transfer and in formation of phototautomer, more stable in non-polar media than anion.

When to add strong acids (H_2SO_4 in water and CF_3COOH in iso-buthylacetate), formation of cationic form ***C*** has been found.

Transformation of all the forms investigated in the ground and excited states is presented on the scheme.



The same investigations of 7-methoxychromones (R = CH_3) showed emission not only of cationic form, but also of another unknown one.